

PROJECT MANAGEMENT
(IT-404)
GROUP: 1
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Short Answers:

1- Name the processes of Project Time Management

ANS:

Project time management is the process of effectively managing the amount of time allocated to various tasks and activities within a project.

It is an essential aspect of project management and requires careful planning, coordination and communication among project stakeholders.

Process:

1. Plan schedule management
2. Define activities
3. Sequence activities
4. Estimate activity resources
5. Estimate activity duration
6. Develop schedule
7. Control schedule

2- Differentiate b/w

<i>Project</i>	<i>Program</i>
• Project is a temporary endeavor with a start and end date it provides a unique output.	• Program consists of multiple projects with each project having its own goal but they have to be managed in such a way that they achieve a program level goal together.
• Project focus on content “What needs to be done”.	• Program focuses on context “How it will be done”.
• Project is short term because one single temporary activity that carries out certain product or service.	• Program is long term because it involves multiple projects in it.
• Project has single functional unit.	• Program has multiple functional units.
• Project is technical in nature.	• Program is strategic in nature.
• Success of a project can	• Success of a program can

- Success of a project can be measured in terms of product quality, timeline, cost effectiveness compliance and degree of customer satisfaction.

- **Example:**
The construction of a new office building. Where project would be define its objectives like a specific timeline, a budget and a scope.

- Success of a program can be measured by the extent to which program meets out the needs and benefits for which it was conducted.

- **Example:**
The development of a new software system for a large organization. It includes multiple projects such as requirements gathering software design, development, testing and implementation.

3- Explain Matrix organization

ANS:

A matrix organization is a type of organizational structure that combines aspects of functional and projectized structures. In a matrix organization, employees are typically grouped by both function (e.g. marketing, engineering, finance) and by project or product teams.

The matrix structure can take different forms, but the most common **types** are:

- **Strong Matrix**
- **Weak Matrix**
- **Balanced Matrix**

The **advantages** of a matrix organization include:

- **Increased flexibility**
- **Improved communication**
- **Better resource utilization**

The **disadvantages** of a matrix organization include:

- **Complex reporting structures**
- **Potential for conflict**
- **Lack of clarity**

Conclusion:

Overall matrix organization can be an effective way to balance the needs of functional departments and project teams, but it requires careful planning and management to be successful.

4- Define Quality by taking example

ANS:

Quality refers to the degree to which a product, service, or process meets or exceeds customer expectations and requirements. A high-quality product or service is one that is reliable, consistent, and performs its intended function well.

For example, let's consider a smartphone. A high-quality smartphone would meet the following criteria:

- **Reliability**
- **Performance**
- **Usability**
- **Durability**
- **Aesthetics**

Conclusion:

Overall, a high-quality smartphone would meet all of these criteria and provide an exceptional user experience that meets or exceeds customer expectation.

5- Explain the concept of Resource Breakdown Structure

ANS:

The Resource Breakdown Structure (RBS) is a hierarchical representation of all the resources that are required to complete a project. It is a component of project management that helps project managers and their teams identify, organize, and manage the resources needed to complete project activities.

- Similar to a work breakdown structure (WBS), which organizes the project tasks into smaller, more manageable components.
- The RBS focuses on the resources required to perform the tasks.
- The RBS includes all the resources needed for the project, such as personnel, equipment, materials, facilities, and any other resources required for the project's successful completion.

For example, at the highest level of the RBS, you may have broad categories of resources such as "Labor," "Equipment," and "Materials."

- " Under the "Labor" category,
- you may have subcategories such as "Project Manager," "Designer," and "Construction Worker."
- Under the "Project Manager" subcategory, you may have specific names of project managers assigned to the project.

- Helps project managers to identify and allocate resources to project activities
- ensure that resources are available when needed and avoiding conflicts.
- Helps in estimating project costs,
- monitoring resource usage, and
- managing resource constraints.

Conclusion:

Overall, the RBS is a useful tool for managing resources in a project, as it provides a structured approach to identifying, organizing, and managing the resources required for project success.

6- Explain Start to start & finish to finish dependencies w.r.t activity sequencing

ANS:

Start-to-Start (SS) and Finish-to-Finish (FF) are two types of dependencies used in activity sequencing, which is the process of determining the order in which project activities should be performed. These dependencies are used to establish the relationship between two or more project activities, which is necessary for creating a project schedule.

Start-to-Start (SS) Dependency:

In a Start-to-Start (SS) dependency, the successor activity cannot begin until the predecessor activity has started. This means that the start of the successor activity is dependent on the start of the predecessor activity. For example, if the predecessor activity is the excavation of a construction site, the successor activity of pouring the foundation cannot begin until the excavation has started.

Finish-to-Finish (FF) Dependency:

In a Finish-to-Finish (FF) dependency, the successor activity cannot finish until the predecessor activity has finished. This means that the finish of the successor activity is dependent on the finish of the predecessor activity. For example, if the predecessor activity is the installation of windows, the successor activity of cleaning the interior of the building cannot be finished until the windows have been installed.

In **summary**, Start-to-Start (SS) and Finish-to-Finish (FF) dependencies are two types of activity sequencing relationships that help determine the order in which project activities should be performed. By using these dependencies, project managers can establish a logical sequence of activities, ensure that all project activities are completed in the correct order, and ultimately help to ensure project success.

Long Answers:

Q#1 what is the concept of Stake holders in project management? Explain its types in details by taking relevant example.

ANS:

In project management, stakeholders are individuals or groups who have an interest or stake in the project's outcome.

These can include individuals or groups within or outside of the organization who are affected by or have an impact on the project.

The concept of stakeholders is critical in project management as it helps to identify the project's key players, their expectations, and how they will be impacted by the project's outcomes.

There are various **types** of stakeholders in project management, and they can be categorized into the following:

- **Internal stakeholders**
- **External stakeholders**
- **Primary stakeholders**
- **Secondary stakeholders**
- **Key stakeholders**

Effective stakeholder management involves engaging with each stakeholder group, understanding their interests and concerns, and developing strategies to manage their expectations.

For example, the project team can communicate regularly with the client to ensure their requirements are met, involve the local community in project planning, and comply with the regulations set by the regulatory bodies. By managing the stakeholders effectively, the project can achieve its objectives while minimizing any negative impact on the stakeholders.

For example, in a construction project, the stakeholders can include the project team, the client and suppliers. Each stakeholder has a different interest and expectation from the project. The project team's interest is to complete the project on time and within budget, while the client's interest is to have a high-quality building that meets their requirements.

Q#2 explain the processes of Project time management in details.

ANS:

Project time management involves analyzing and developing a schedule and timeline for project completion.

Six Main Processes In Project Time Management

1. Creating A Plan For A Project:

“The document that describes how the project will be executed out, monitored, controlled, and closed.”

- Outlines Goal/Scope Of project
- Provides an Official point to the Team, company and Stakeholders
- Detail planning will save alot of time as Project processes

2. Project Schedule Management Process:

“Project Scheduling is more than just a simple calendar”.

- Shows the Timeline for the project
- shows organizational resources for each task
- Provides the critical information to the team
- It needs to be comprehensive and easy to understand.

3. Define The Project's Activities:

“A list of the tasks that need to be done in order to finish your project”

- For complex Projects, organize these tasks into WBS/ a charts that helps to visualize projects and their tasks.

4. Determine Dependencies:

"Figuring out which activity depends on whom"

- It's important to define all of your project's dependencies correctly so you can make a good schedule and avoid delays.

E.g. If you're building a house you can't put the roof until the frame is done.

5. Sequence Activities:

"Putting all the different activities in an order"

- Project activities should be done in an order to make the best flow.

6. Estimate Durations And Resources:

It can be done in few different ways:

- Use data from similar jobs that have already been done.
- Make estimate by taking the average of the best, worst, and most likely cases
- Estimate how many resources each project activity will need.

Q#3 Explain at least two techniques of cost estimation in details by taking relevant examples.

ANS:

What is cost estimation?

It's the process of estimating all of the cost associated with completing a project within scope and according to its timeline.

Cost estimation techniques/methods:

- Analogous Estimation
- Parametric Estimation
- Bottom up Estimation
- Three point Estimation

Bottom up Estimation:

- A larger project is broken down into number of smaller components.
- The project manager then estimate cost specifically for each of these smaller work packages.

For example, if you are building a house, you could break down the project into smaller tasks or work packages, such as excavating the foundation, pouring the concrete, framing the walls, installing the plumbing, and so on.

- You would then estimate the cost of each of these tasks based on the required resources, materials, labor, and equipment.
- The estimated costs of the individual tasks are then added up to arrive at the total cost estimate for the project.
- Bottom-up estimating is a time-consuming process that requires a lot of detailed information, but it provides a more accurate cost estimate than other techniques.
- It is particularly useful for complex projects with many different tasks or work packages.

Analogous Estimation:

- . This technique is based on the assumption that the cost of a new project is similar to the cost of a previous project that has similar characteristics, such as scope, size, complexity, and technology.

For example, if you are planning a new software development project, you could use historical data from similar software development projects to estimate the cost of your project.

You would need to identify projects that are similar in terms of scope, size, complexity, and technology, and use the cost data from these projects to estimate the cost of your project.

You could adjust the estimates based on the differences between your project and the previous projects.

- Analogous estimating is a quick and easy technique
- can be useful when detailed information is not available or
- when time is limited.
- it can be less accurate than other techniques,
- particularly if the previous projects are not truly similar to the new project.